QUALYXUS

Air-Gapped Edge AI for Visual Inspection & Process-Quality Intelligence

- **Startup Name:** Qualyxus
- **Founder & Lead Architect:** Utkarsh Sengar (B.Tech CSE, IIT Dharwad)
- **Prepared for:** dhaRti Foundation, IIT Dharwad · November 2026

> **We solve** late-stage scrap and escaped defects **by providing** an on-premise edge-AI platform that inspects every part and links each defect to the machine cycle that produced it, **to help** automotive, aerospace & defense, and industrial manufacturers **trace** defects to process drift and machine wear, so causes get fixed sooner, with all data staying inside the plant.

Compelling Problem Statement

The Problem: Quality and Machine Data Live in Separate Silos

Plants run two disconnected systems, so the defect and its cause are rarely seen together:

1. Machine Vision (Post-Mortem Sorting)

- Inspects parts **after** they are made, so scrap is caught only after material, machine hours and labour are spent.
- Rule-based setups need re-programming for every new part and struggle with variable cosmetic defects.
- Says pass or fail, but not **why**: blind to the machine conditions that caused the defect.

2. Machine Telemetry (Blind to Outcomes)

- Monitors temperatures, pressures, spindle load and vibration to prevent breakdowns.
- Never learns which parts came out bad, so it cannot tell which signal patterns cause defects.
- A machine can stay inside every alarm limit while slowly drifting into producing scrap.

Target Customer Profile: Plant Heads, VPs of Quality and Maintenance Heads running automated machining, moulding, casting, stamping, welding and assembly lines in automotive, aerospace & defense, and industrial manufacturing.

Solution Offered & Innovation Proposed

The Causal Bridge: Every Defect Linked to Its Machine Cycle

Qualyxus fuses camera inspection with machine-cycle data on one on-premise edge box, so quality and maintenance teams see the same cause.

Camera + Machine Data
Industrial cameras; Modbus TCP / OPC UA

On-Prem Edge Engine
Inspect, link to cycle, score quality & wear

Advisory PLC Signal
One verdict per part; plant logic decides

Core Deep-Tech Innovations

1 Learns from Good Parts

Anomaly detection (Dinomaly / PatchCore on DINOv2) trains on several hundred good parts, with no defect library needed. AUROC 0.95–0.99 on 4 categories of the public VisA benchmark.

2 Calibrated False Rejects

Reject thresholds are set by statistical (conformal) calibration with a stated confidence bound, per part across all cameras, so the plant knows its false-reject rate before go-live.

3 Root Cause & Machine Health

Each part is linked to the cycle that made it. A wear index per component tracks drift (heater bands, check ring, cooling), with approval-gated fixes that cut scrap 79% in simulation.

4 Air-Gapped by Design

Runs fully on-premise on an NVIDIA GPU, verified offline with no downloads. Shadow mode, one-command rollback and a tamper-evident audit log built in.

Built & Measured

Working Prototype, Pre-Revenue: What the Platform Already Does

0.95–0.99

Anomaly-detection AUROC on 4 categories of the public VisA benchmark (previous baseline: 0.56–0.96). With thresholds set for low false rejects, it caught 35–90% of defects, depending on the part.

~1%

False rejects per part, calibrated across a simulated six-camera station: 2 of 400 unseen good parts, with all 4 defect types caught (synthetic).

100% → 0%

False rejects on parts rotated 8° after automatic part registration, with all defects still caught; 15% at 15° (synthetic).

0.83

Mask IoU for defect segmentation (RF-DETR-Seg): 100% recall and defect-type accuracy, 16.5 ms per image on a laptop GPU (synthetic).

~4 s

To promote a new model live with no downtime; 98.5% shadow-mode agreement in closed-loop simulation.

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Automated tests passing, including forced failures: dead cameras, PLC outage, power cuts. End-to-end offline runs keep the audit log intact (simulation).

Results on the public VisA benchmark and on our synthetic part and process simulator. Nothing has run on a real PLC, camera, Jetson or factory line yet: measuring that is the goal of the paid pilots.

Competition & Comparison Matrix

Where Qualyxus Fits

	Legacy Vision (Cognex / Keyence)	AI Vision Competitors (SwitchOn)	Heavy Automation (Siemens / Bosch)	QUALYXUS PLATFORM
Operational Method	Post-Mortem Sorter (catches scrap late)	Primarily Visual Inspection	Heavy Physical Sensor Wiring	**Inspection + Process Drift & Wear Tracking**
Data Underpinnings	Rule-Based Image Tools (+ add-on deep learning)	Deep-Learning Image Inspection	Machine Telemetry Logs	**Images Linked to Machine-Cycle Data**
Deployment Time	Weeks of Integrator Programming per Part	Under 45 Min per New SKU (Vendor Claim)	Multi-Month Projects	**Target: 1–2 Day Install (No Machine Rewiring)**
Safety Compliance	N/A (Post-Line)	N/A (Visual Check)	Changes May Need Safety Re-validation	**Advisory Only (No Safety-Function Changes)**
Budget Captured	Quality Control Only	Primarily Quality Control	Maintenance Only	**Dual Budget: Quality + Maintenance**
Typical Cost to Plant (indicative)	₹1.2L – ₹7.5L per smart camera + integration (one-time)	Subscription (pricing not public)	₹50L+ CapEx projects	**₹2–2.5L setup + ₹20k / month per line**

Cognex In-Sight 2000 to 8000-series dealer listings in India (ex-tax, lens and integration extra). Ranges are indicative. Others also link quality to process data (e.g. Kistler ComoNeo, Eigen, Sciemetric); our difference is learning from every inspected part, on-premise, at retrofit cost.

Market Research & Opportunity

TAM / SAM / SOM — Bottom-Up Sizing

$45B

TAM (Total Addressable Market)

~₹3.9 lakh crore — Global AI visual inspection (~$31B) + predictive maintenance (~$14B), 2025 estimates.[1]

$590M

SAM (Serviceable Addressable Market)

~₹5,100 Cr / yr — ~1.7 lakh inspection points on machining, moulding, casting & forging, stamping, welding and assembly lines (~1.3 lakh India, ~0.4 lakh SE Asia) × ~₹3L blended contract.[2]

$5.5M

SOM (Serviceable Obtainable Market)

~₹48 Cr ARR — 2,000 lines in Indian industrial corridors (Pune, Chennai, Bengaluru, NCR, Hyderabad, Hubballi–Belagavi) × ₹2.4L list / yr. 5-year target (~1.5% of India); exports additive.

Market Tailwinds & Growth Drivers

Auto-Component Boom

₹7.6 lakh crore turnover in FY26, up 12.7% year on year, as global OEMs shift sourcing to India.[3]

India PLI Expansion

₹1.97 lakh crore PLI outlay across 14 sectors, incl. auto components, ACC batteries and electronics.[4]

Large Installed Base

₹31,781 Cr of machine tools consumed in FY25[5], ~5,000 foundries and 500+ forging units[6]: retrofit beats ₹50L+ replacement.

[1] MarketsandMarkets, Market Research Future; IMARC. [2] Qualyxus estimate. [3] ACMA, Jul 2026. [4] PIB, Nov 2024. [5] IMTMA (provisional). [6] Foundry Informatics Centre; AIFI. ₹87 = $1.

Business Model & Unit Economics

Hybrid Hardware-Enabled SaaS

How we land: 60-day paid pilot beside manual inspectors (₹1L, credited on conversion), then 5–8 lines per plant.

1. One-Time Setup & Hardware: ₹2.0L – ₹2.5L per line

- Single-camera station: NVIDIA Jetson Orin-class edge computer, industrial camera + lens, lighting, enclosure, target 1–2 day install.
- Multi-camera "All-Faces" station: ₹5L – ₹7L.
- Target hardware gross margin: **20–30%**

2. Software Licence: ₹20,000 / month per line

- ₹2.4L / year, billed annually: models, retraining, updates, dashboard, on-site support.
- Costs no more than one inspector (₹20–35k / month loaded), yet covers all three shifts.
- Target software gross margin: **80%+**

Customer Payback (per line)

Metric	Value
Annual value delivered	₹6L – ₹15L typical; up to ₹25L on high-value parts
Year-1 cost to customer	₹4.4L – ₹4.9L (hardware + licence)
Payback period	4 – 10 months
Year-2+ return on licence	2.5× – 6×

Qualyxus Unit Economics (targets)

Metric	Target
Lines per plant (within 24 months)	5 – 8
Customer acquisition cost per plant	₹5L – ₹8L (pilot + field sales)
Gross profit per plant at 6 lines	~₹11.5L / year
CAC payback	5 – 9 months

All figures are planning assumptions to be validated in paid pilots.

Team & Advisory Details

Leadership, Cap Table & Strategic Advisors

Founder & Core Architect

Utkarsh Sengar (B.Tech CSE, IIT Dharwad)

- Built the full platform: good-parts-only anomaly detection, defect segmentation, part-to-cycle linkage and risk models, edge deployment and model operations.
- LinkedIn: linkedin.com/in/utkarsh-sengar

Cap Table Status

100% Founder Owned (Utkarsh Sengar)

Company incorporation planned with dhaRti support.

Strategic Advisory Board

Dr. Rakesh Kumar

VP R&D, Rossell Techsys (ex-Moog, ex-Indian Navy).

Shankar Lal

Discipline Chief, Cyient (21 years at HAL).

Ramesh Rana

Operations Lead, SIG SAUER.

How dhaRti Can Help

What I Need to Reach the First Paid Pilots

1 Incubation & Lab Space

A campus desk, a lab bench for a test inspection rig (camera, lighting, sample parts) and GPU compute time.

Unlocks: Real-hardware testing before plant pilots.

2 Seed Support: ₹5L to Start

Up to ₹15L in milestone-based tranches (next slide), from dhaRti directly or through NIDHI-PRAYAS / Startup India Seed Fund with dhaRti as host.

Unlocks: First 3 pilot stations and Jetson benchmarks.

3 Pilot Introductions

Introductions to 3–5 plants in the Hubballi–Dharwad–Belagavi cluster: foundries, machining shops and aerospace suppliers.

Unlocks: First paid 60-day pilots, close to campus.

4 Mentors

A manufacturing-quality expert, an industrial-sales mentor and an IP advisor.

Unlocks: Faster, better-aimed pilots and pricing.

5 IP & Company Setup

Patent assessment and filing for our part-to-cycle linking and calibration methods; incorporation and compliance guidance.

Unlocks: Defensible IP before wider rollout.

6 Investor Access

Introductions to angels and funds for the ₹2 Cr pre-seed once pilots are live.

Unlocks: The ₹2 Cr pre-seed after pilots show results.

In return: an IIT Dharwad deep-tech startup in manufacturing AI, with measured results from real plants within 9 months of incubation and monthly progress reports to dhaRti.

Seed Support Request

₹5 Lakh to Start; Up to ₹15 Lakh over 9 Months as Milestones Are Met

Use of Funds

Item	Amount
3 pilot inspection stations (camera, lighting, Jetson)	₹5.0L
Lab test rig + Jetson development kit	₹1.5L
Engineering intern / junior engineer, 6 months	₹3.0L
Plant installs and travel	₹1.5L
Incorporation, DPIIT recognition, provisional patent	₹1.5L
Compute, software and tools	₹1.0L
Contingency	₹1.5L
Total	₹15.0L

Released Only as Milestones Are Met

Tranche 1 · ₹5L · Month 0

Test rig, first station, company setup and patent filing.
Milestone (month 3): rig live; Jetson speed and accuracy benchmarks published.

Tranche 2 · ₹5L · Month 3

Two more stations and the intern's first 3 months.
Milestone (month 6): 3 pilot stations installed in nearby plants.

Tranche 3 · ₹5L · Month 6

Intern months 4–6, plant installs and travel, contingency.
Milestone (month 9): measured pilot results and the first paid conversion.

dhaRti's risk is capped at ₹5L at a time: each tranche follows a met milestone. Can be funded directly or through NIDHI-PRAYAS / SISFS with dhaRti as host incubator.

QUALYXUS

Edge AI that traces every defect to its cause

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Demo
Live demo and code walkthrough available on request